

Sampling variability

Project: Reproducible AI Research Study

1. Your mission

Your learning target

I can explain the mechanism and terminology of sampling variability.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of sampling variability

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Confounders and controls. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain sampling variability using the records and boundaries of this fictional investigation.

2. Notice the evidence

Sampling variability means different eligible samples can give different estimates even under the same population. Separate this from procedure changes.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 without replacement. Their means are 1, 2, 2, 2, 2, 3.

Sampling variability — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Find the mean of sample means and the chance of mean 2 under uniform selection of those six samples.

Check or calculation: Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 without replacement.

Conclusion and limit: $\text{Mean} = (1+2+2+2+2+3)/6 = 2$; mean 2 occurs $4/6 = 2/3$. The population mean is 2, while individual samples differ.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Find the mean of sample means and the chance of mean2 under uniform selection of those six samples. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Evaluate the exact same fixed four cases twice with deterministic predictions.

Prediction: The counts should agree unless some condition, data or scoring changed.

Reason: The sampling explanation no longer applies when the records are identical.

5. Words that unlock the idea

Word · Meaning

Word: Sampling variability

Meaning: Changes in estimates across eligible samples

Word: Sampling distribution

Meaning: The distribution of an estimate under a sampling procedure

Word: Without replacement

Meaning: Selecting distinct units within a sample

Use two terms to explain a step in this week's example.

6. Follow a worked example

Find the mean of sample means and the chance of mean2 under uniform selection of those six samples.

Mean=(1+2+2+2+2+3)/6=2; mean2 occurs4/6=2/3. The population mean is2, while individual samples differ.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program uses fictional values and the Python standard library; it performs no external operations.

```
from itertools import combinations

def sample_means(population, size):
    if type(size) is not int or not 1 <= size <= len(population):
        raise ValueError('sample size')

    return [sum((population[i] for i in indices)) / size for indices in
combinations(range(len(population)), size)]

print(sample_means([1, 1, 3, 3], 2))
```

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sampling variability

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

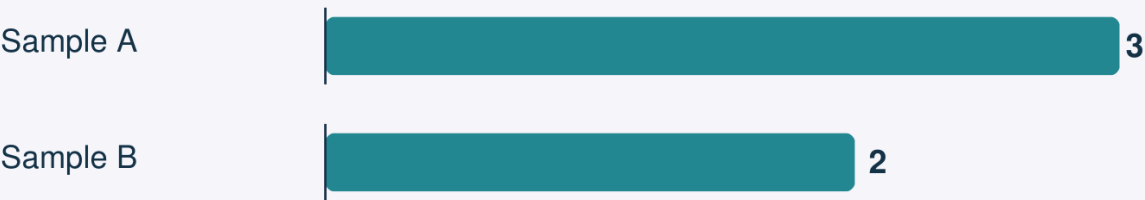
The facts we will use

Sample A has 3 correct of 4; sample B has 2 correct of 4. The model and rubric are unchanged.

AI ACADEMY / AI-7/11

Two distinct samples, fixed model

correct cases out of 4 in each sample



Bar length shows the supplied value; the marked baseline is zero.

Sampling variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep model fixed

No training or threshold change occurs between samples.



2. Compute observed rates

A is 75%; B is 50%, differing by 25 percentage points.



3. Consider sampling

Different included cases can change a small-sample rate.

CHANGE ONE THING

Evaluate the exact same fixed four cases twice with deterministic predictions.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Keep model fixed

What happens and why: No training or threshold change occurs between samples.

Step: 2. Compute observed rates

What happens and why: A is 75%; B is 50%, differing by 25 percentage points.

Step: 3. Consider sampling

What happens and why: Different included cases can change a small-sample rate.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

What this example does not establish

These two samples do not supply a probability model or confidence interval; none is invented.

Read the labelled picture: Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

Change one thing: Evaluate the exact same fixed four cases twice with deterministic predictions. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

This finite example is not a general normality guarantee or a model of a biased convenience sample.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample. Explain how omitting it could change a reader's interpretation, and write an

accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Sampling variability”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Sampling variability. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Find the mean of sample means and the chance of mean2 under uniform selection of those six samples. Show the intermediate steps and give the conclusion with its scope.

```
from itertools import combinations

def sample_means(population, size):
    if type(size) is not int or not 1 <= size <= len(population):
        raise ValueError('sample size')

    return [sum((population[i] for i in indices)) / size for indices in
```

```
combinations(range(len(population)), size)]

print(sample_means([1, 1, 3, 3], 2))
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: This finite example is not a general normality guarantee or a model of a

biased convenience sample. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled sampling variability evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of sampling variability. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Sampling variability — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Find the mean of sample means and the chance of mean2 under uniform selection of those six samples.

Check or calculation: Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 without replacement.

Conclusion and limit: $\text{Mean} = (1+2+2+2+2+3)/6 = 2$; mean2 occurs $4/6 = 2/3$. The population mean is 2, while individual samples differ.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

Population 0, 0, 4, 4 with distinct IDs; enumerate size 2 sample means and their average.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Population0, 0, 4, 4 with distinct IDs; enumerate size2 sample means and their average.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of sampling variability

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of sampling variability

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Different included cases can change a small-sample rate.

1. Consider sampling
2. Keep model fixed

3. Compute observed rates

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Consider sampling. Different included cases can change a small-sample rate. Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

Practice 2: explain the deciding evidence

Changed-case practice: Evaluate the exact same fixed four cases twice with deterministic predictions. Which conclusion is supported by the supplied facts?

1. The counts should agree unless some condition, data or scoring changed.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The counts should agree unless some condition, data or scoring changed. The sampling explanation no longer applies when the records are identical.

Repair a missing building block

Review Confounders and controls (ai-7/10). Goal: Explain the mechanism and terminology of confounders and controls. Short practice: Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?. Return to this week and explain what you changed.

Review Baselines and fair comparisons (ai-7/8). Goal: Explain the mechanism and terminology of baselines and fair comparisons. Short practice: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple baseline?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the mechanism and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets. Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.